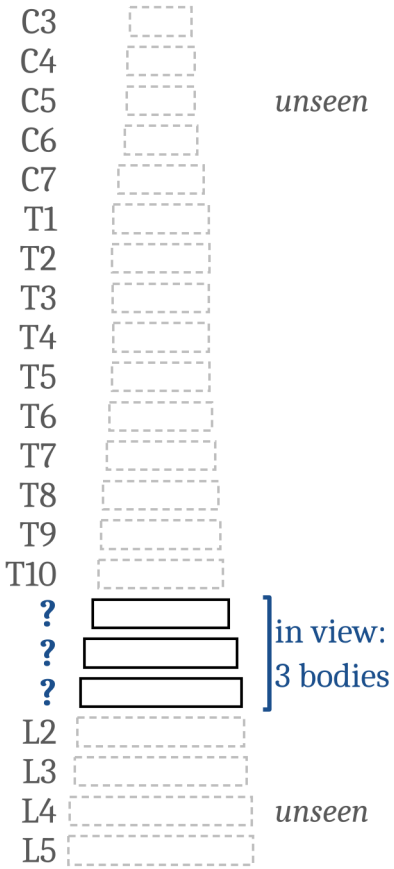
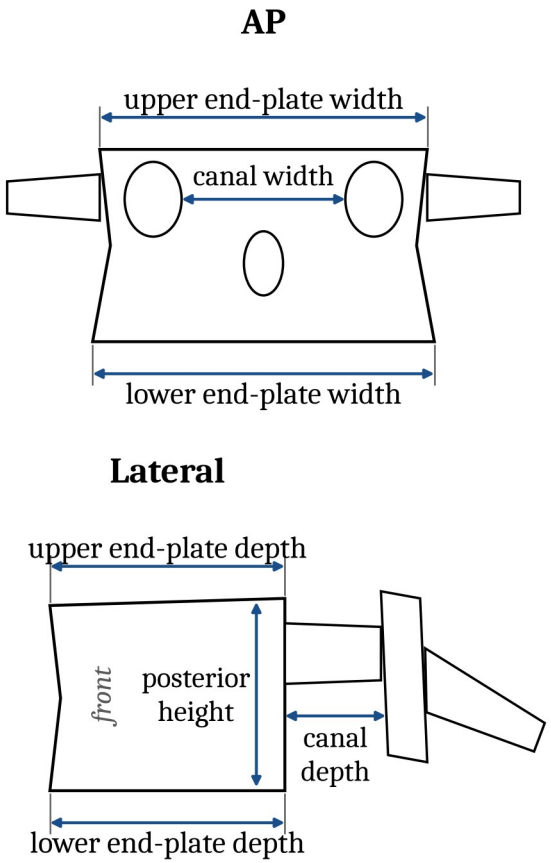
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Evolution

Can the bodies in view name a vertebra?





Gregory Schwing, MD, PhD
General Surgery (Wayne State University School of Medicine / Detroit Medical Center)

A view shows a few consecutive vertebral bodies, often with no C2 or sacrum to count from.

Seven measures per body. Which levels are in view?

2,000 synthetic columns drawn from Panjabi's published means and spreads; every field of every column scored, so most rates below are over tens of thousands of fields.

Synthetic columns from Panjabi's published tables (Spine 1991;16:861, 888; 1992;17:299); no patient data.

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Evolution

More bodies in view, more certainty about the level

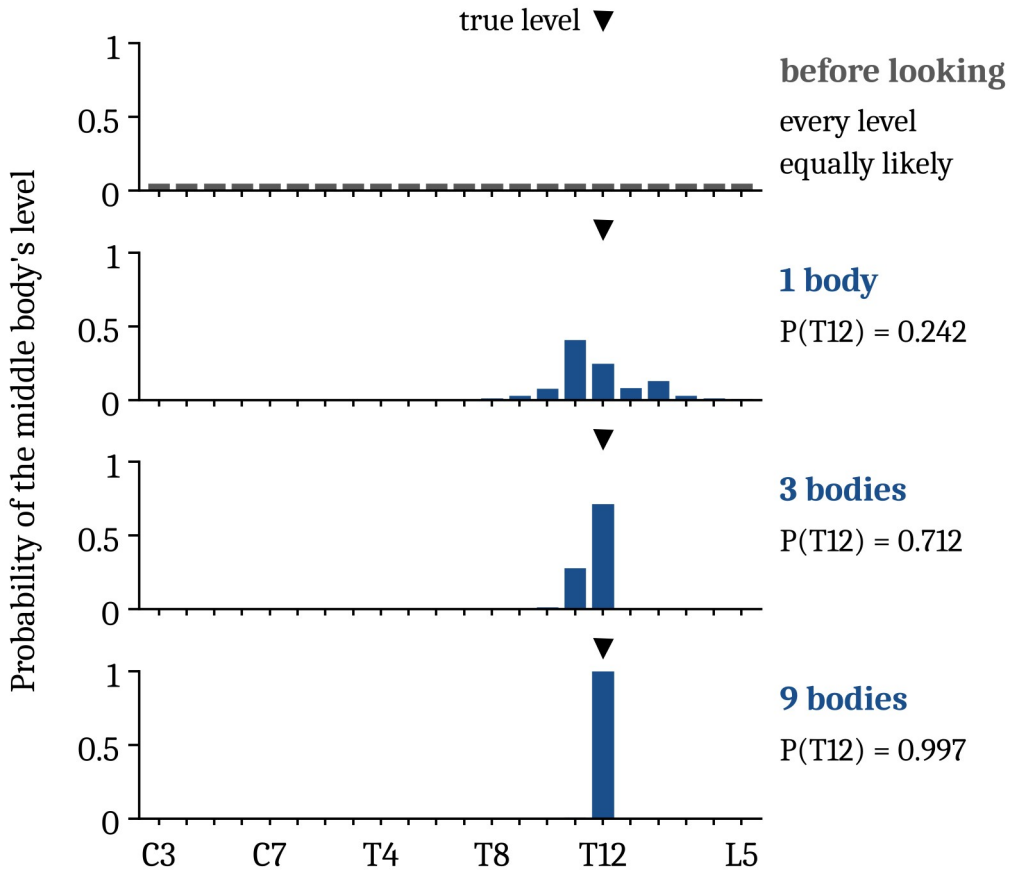
Fields centred on T12 in synthetic columns, named exactly:

39%
from one body

73%
from three bodies

95%
from nine bodies

No formula or network on these measures names more fields correctly on average.



Synthetic columns from Panjabi's published tables (Spine 1991;16:861, 888; 1992;17:299); no patient data.

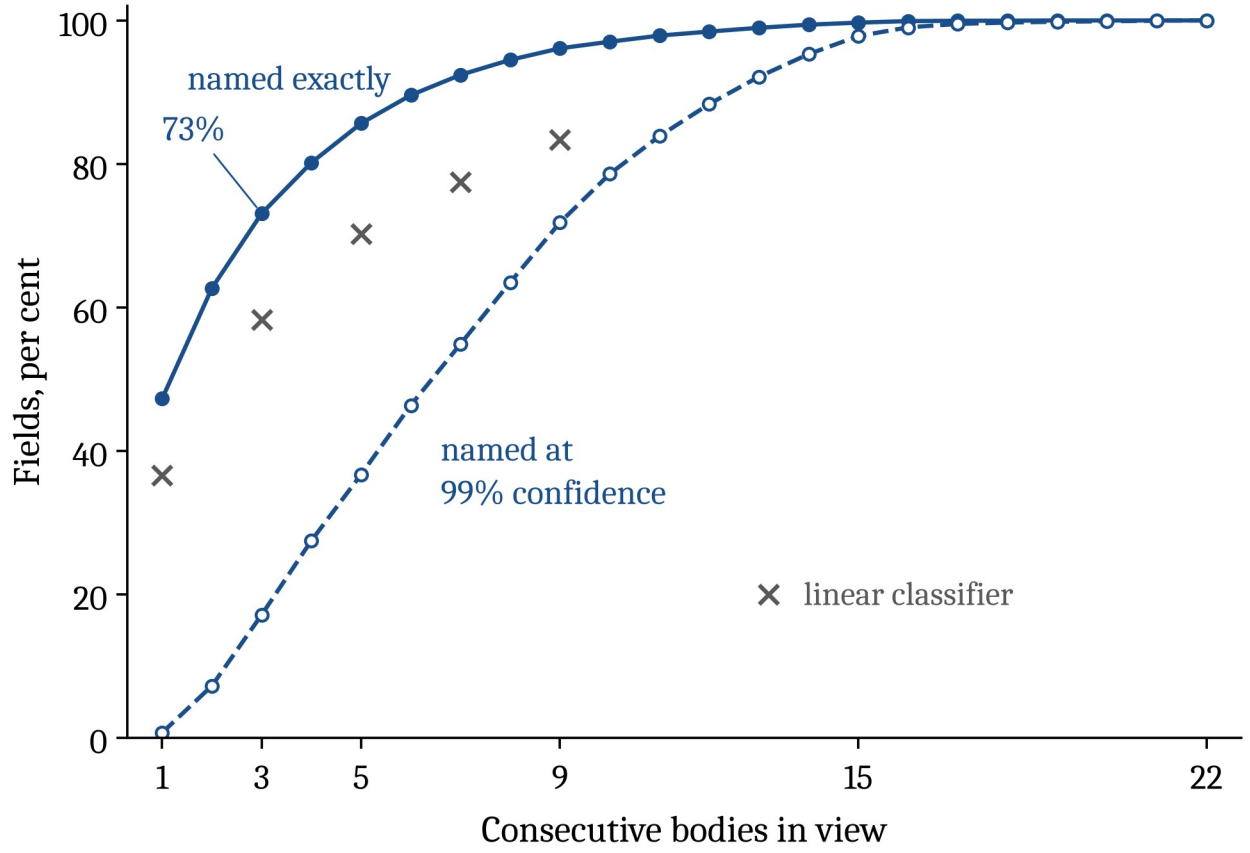
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Three bodies name 73%; when sure, 99.8% are right



73%

of three-body fields named exactly (a linear classifier: 58.2%; a fitted network: 71.6%)

99.8%

of three-body names given at 99% confidence right (15 wrong of 6,848; named for 17% of fields)

96%

named exactly from nine bodies

Named only at 99% confidence, it was right 99.7% or more of the time at every number of bodies in view, from 1 to 22, on synthetic columns.

Synthetic columns from Panjabi's published tables (Spine 1991;16:861, 888; 1992;17:299); no patient data.

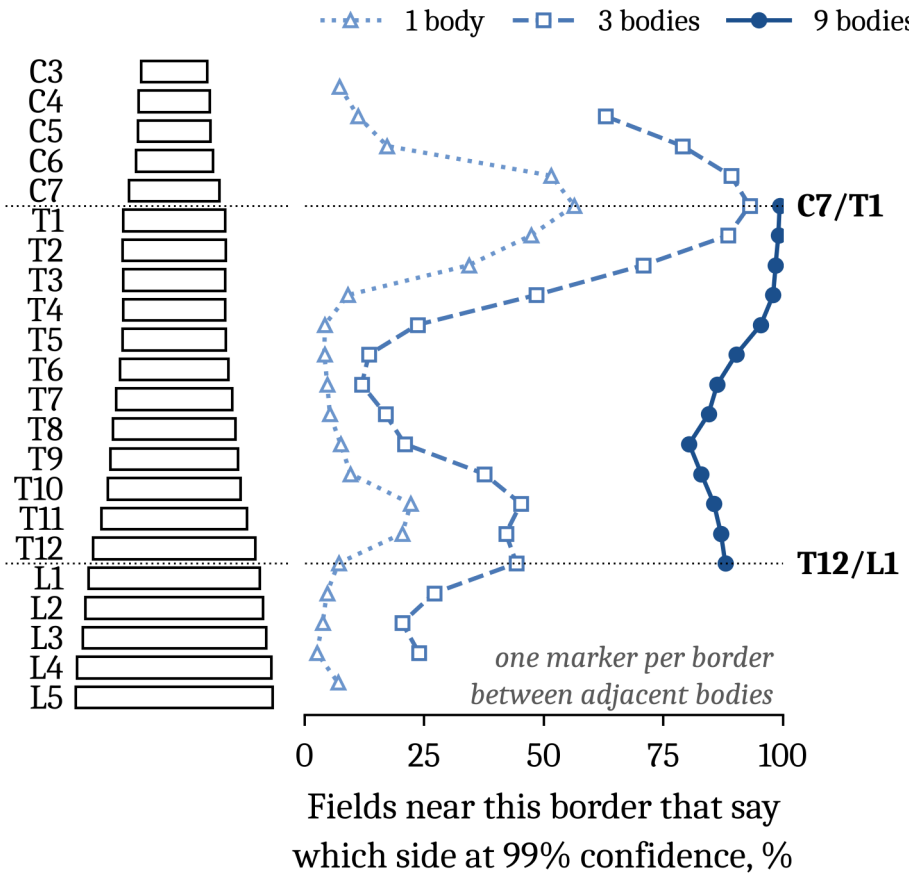
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Evolution

One body tells its region and its side of C7/T1, rarely of T12/L1



Claims made at 99% confidence are right at least 99% of the time over all fields (T12/L1 from one body: 99.96%, 13 wrong of 30,943). The rates below are for bodies within two levels of a border, chosen by their true level, where a claim can fall short.

56%

of single bodies within two levels of C7/T1 say which side; right 99.8% (7 wrong of 4,514)

7%

of single bodies within two levels of T12/L1 say which side; right **97.9% (12 wrong of 568), short of 99%**

61%

of single bodies say their region; right 99.9% (21 wrong of 26,786), only 2 of them far from both junctions

88%

of fields with middle body T11 to L1 say which side of T12/L1 from nine bodies (1 wrong of 5,279)

Synthetic columns from Panjabi's published tables (Spine 1991;16:861, 888; 1992;17:299); no patient data.

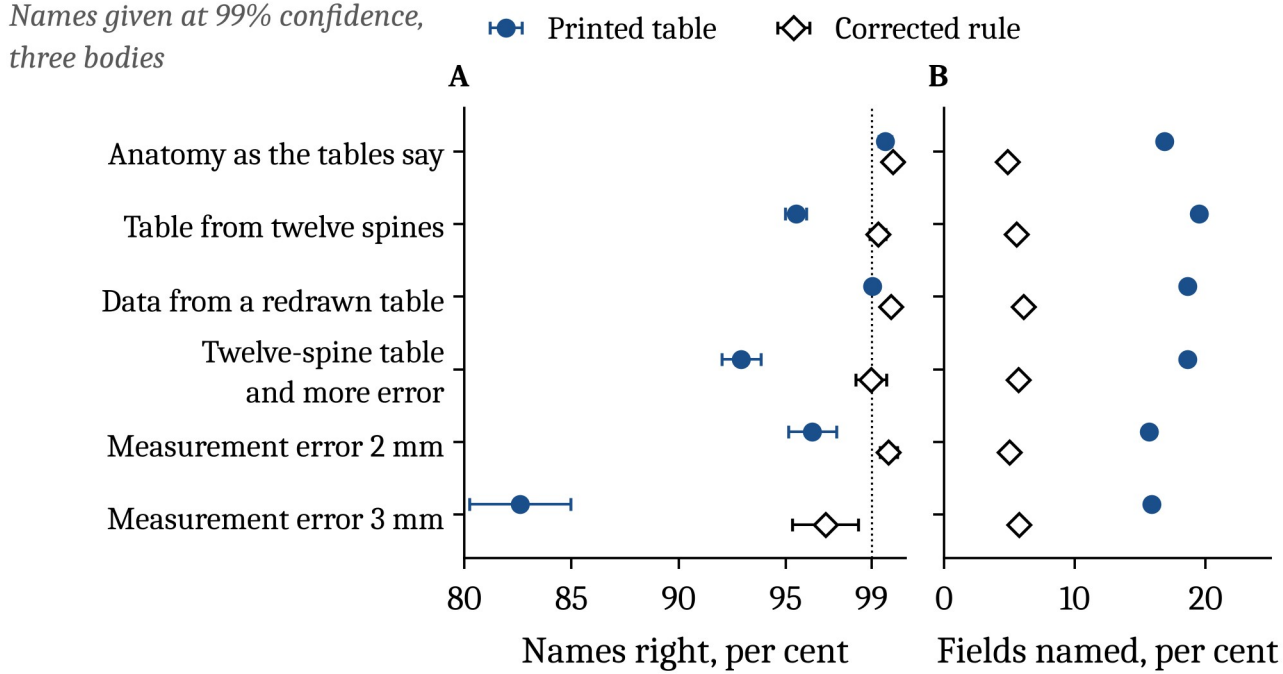
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Evolution

The weak point is the table, not the mathematics



95.5%

of three-body names at 99% confidence right when the rule's table is estimated from 12 spines, as Panjabi's tables are

99.3%

with a corrected rule, its one setting chosen on a separate pilot, but 5.5% of fields named, not 19.5%, and lumbar names right 96.7% (8 wrong of 246)

Next: test the rule on real full-spine CT.

Synthetic columns from Panjabi's published tables (Spine 1991;16:861, 888; 1992;17:299); no patient data.

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